

USN

--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Sixth Semester B.E. (Biotechnology) (Credit System) Degree Examinations****Supplementary Examinations - December 2025****21BTE101/20BTC101 - ADVANCED PLANT TISSUE AND ORGAN CULTURE**

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **One full** question from **Unit – I** and **Two full** questions from **Unit – II & Unit – III** each.*

Unit – I		Marks	BT*	CO*	PO*
1. a)	Discuss on micropropagation which helps in large scale clonal propagation.	10	L*2	1	1
b)	Comment on cybridization. With examples, discuss selection process of somatic hybrids.	10	L2	1	1
2. a)	Another culture is responsible for double haploid production. Discuss.	10	L2	1	1
b)	Discuss on embryo culture and give an account on its importance.	10	L2	1	1
Unit – II					
3. a)	With a case study, explain about Agrobacterium mediated transformation through protoplast culture.	10	L2	2	2
b)	What is the need for marker free transgenics? Explain various approaches involved in the production of marker free transgenics.	10	L2	3	2
4. a)	Describe how cointegrate vectors are created and transformed plant cells selected.	10	L2	2	2
b)	How RAPD helps to distinguish monomorphism and polymorphism? Discuss.	10	L2	3	2
5. a)	Comment on the methods involved in chloroplast transformation.	10	L2	2	1
b)	Explain about the use of SCAR on molecular markers.	10	L2	3	2
Unit – III					
6. a)	Give examples of various genes and proteins involved in drought response.	10	L2	4	1
b)	Discuss on genetic modification of plants with delayed fruit ripening.	10	L2	5	2
7. a)	Illustrate pathway involved in glyphosate and glufosinate action on weeds.	10	L2	4	1
b)	How non Bt mode of developing insect resistance in plants are done? Discuss.	10	L2	5	2
8. a)	Explain different molecular mechanisms for resistance in plants against nematodes.	10	L2	4	1
b)	Discuss various approaches involved in the development of virus resistant plants.	10	L2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
